L1.1b Transformations

Work in your group. Check your answers against the Quarto tonight.

1 · Transformations

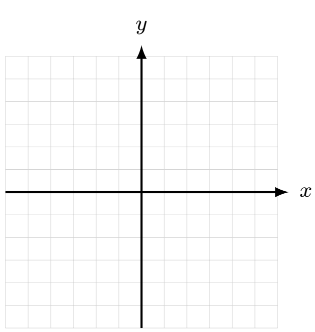
1. Classify — do not compute. Say the move as a direction, never as "left" or "right."

	outside / inside	**moves D or R**	**the move, as a direction**
$y = f(x) - 4$			
$y = f(x - 4)$			
$y = 3f(x)$			
$y = f(3x)$			
$y = -f(x)$			
$y = f(-x)$			
$y = \lvert f(x) \rvert$			
$y = f(\lvert x \rvert)$			

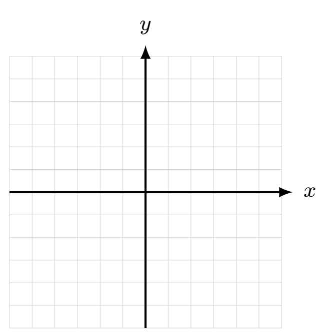
2. Read the moves off the equation, then sketch. Name the parent, state the moves in words, sketch, and give D and R in both notations. **No T-charts.**

> **Stop.** Before you sketch — what does something *outside* the $f(x)$ move? What does something *inside* move, and which way?

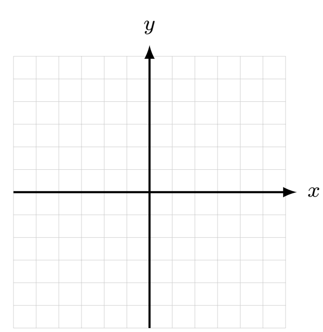
(a) $y = \sqrt{x} - 2$



(b) $y = \sqrt{-x}$



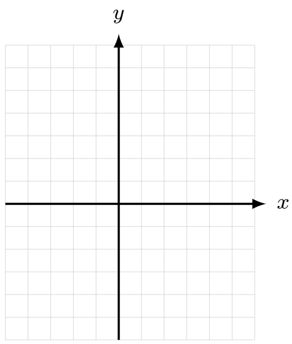
(c) $y = -(x + 2)^2 + 1$



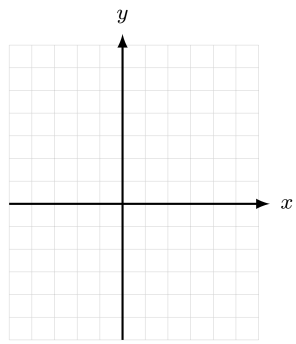
3. Backward. Write the equation: x^2 stretched vertically by 3, reflected over the x-axis, slid 4 in the $-x$ direction and 2 in the $+y$ direction.

4. The absolute value pair. $f(x) = x^2 - 2x - 3$. Factor it, then sketch $|f(x)|$ and $f(|x|)$ on separate axes. Give R for each, both notations.

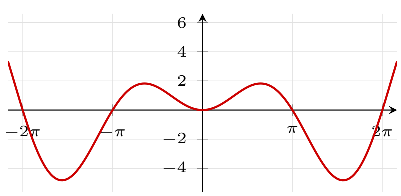
$|f(x)|$



$f(|x|)$

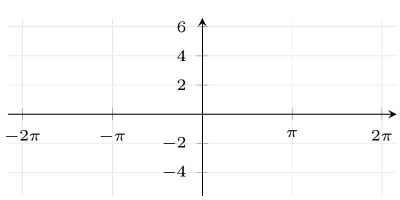


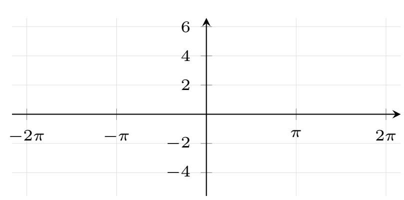
5. $f(x) = x \sin x$, graphed below.



(a) Is f even, odd, or neither? Show it algebraically.

(b) Sketch $f(2x)$ on the left grid and $f(|x|)$ on the right.





(c) One of those two sketches came out identical to f. Which, and what does (a) have to do with it?

2 · Combining functions

6. Complete the square, then read the shifts. $f(x) = x^2 - 8x + 11$. Give the vertex form, the vertex, and the two moves off x^2 — as directions.

7. $f(x) = \dfrac{1}{x^2}$ and $g(x) = \cos x$. Find each function **and its domain**.

(a) $(f+g)(x)$ (b) $\left(\dfrac{f}{g}\right)(x)$ (c) $\left(\dfrac{g}{f}\right)(x)$

3 · Composition and diffnifod

8. Order matters. $f(x) = \sqrt{x}$ and $g(x) = x^2 - 9$. Find each composite and its domain, both notations.

Stop. Write the domain rule down before you touch these — both halves, and what each half is checking.

(a) $g(f(x))$ (b) $f(g(x))$

9. $f(x) = \dfrac{1}{x-2}$ and $g(x) = \sqrt{x}$. Find $f(g(x))$ and its domain.

10. $f(x) = \dfrac{1}{x^2}$ and $g(x) = \cos x$ again. Find $(f \circ g)(x)$ and $(g \circ f)(x)$, each with its domain.

At the boards

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11. BOARDS $f(x) = \ln(x-4)$ and $g(x) = \sqrt{x+7}$. Find $(f \circ g)(x)$ and its domain.

Then do the same for $F(x) = \sqrt{1-x}$ and $G(x) = x^2 + 10$: find $F(G(x))$ and its domain.

One of these two composites does not exist. Which one — and which half of the rule told you?